MIL-C-51449(EA)
30 June 1978
(See section 6)

MILITARY SPECIFICATION

CANISTER, 2.75 WP

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of filled canister.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus (WP).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.

FSC 1340

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

D90-1-302 - Canister, 2.75 WP.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D90-1-302 shall conform to the specification listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the canister shall conform to the specification and drawings listed on Drawing D90-1-302 and subsidiary drawings.

3.2 Filling and assembly.

3.2.1 Filling. The canister, clean and free of foreign matter, shall be filled, using a dry fill method, with molten WP, conforming to MIL-C-215 (see 6.3 and 6.4), at a temperature of 150° ± 10° F (65.5° C ± 5.6° C). When water is in contact with the WP, it shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. After test for leakage, the WP shall be solidified with the canister in a vertical position and filling port up.

3.2.2 Filling height. There shall be a 7 ± 1 percent void, by total volume, existing above the surface of the liquid in the filled and closed canister. The canister shall be radiographed in a vertical, filling port up position while the temperature of the WP is 150° ± 10° F (6.56° C ± 5.6° C) when tested as specified in 4.2.4.1 to determine filling height (see 6.3).

3.2.3 Assembly. The canister shall be assembled and closed as shown on Drawing D90-1-302 (see 6.3).

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3.3 Cleaning. The exterior of the closed canister, the interior of the burster well, and the burster well threads shall be thoroughly cleaned to remove all grease, oil, and WP contamination (see 6.5).

3.4 Leakage. All joints shall be WP wetted prior to oven testing. The closed and cleaned canister shall not leak when maintained at 160 ± 5° F (71.1° C ± 2.8° C) for a minimum period of 8 hours and tested as specified in 4.2.4.2. Alternate method: the canister shall not leak when maintained at 207° to 212° F (97.2° C to 100° C) for a minimum period of 15 minutes and tested as specified in 4.2.4.2.

3.5 Workmanship. The canister shall be free from foreign matter (oil, grease, etc.) and damage such as dents, cracks, and deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of section 3 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. A lot shall consist of like components or subassemblies of one interfix number from one manufacturer, from the same materials and under essentially the same manufacturing conditions.

4.2.2 Sampling. Sampling for examination and tests shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection.

4.2.3.1 For examination and test. Sample canisters shall be examined and tested in accordance with classifications of defects and MIL-STD-105.

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4.2.3.2 For critical defects. Each canister shall be inspected for the critical characteristics listed in the classification of defects.

4.2.3.3 Classification of defects. See page 5.

4.2.4 Tests.

4.2.4.1 Radiographing. The filled and closed canister shall be radiographed while the WP is still molten as specified with the canister in a vertical filling position filling port up and inspected to assure proper filling height in accordance with Drawing D90-1-302.

4.2.4.2 Leakage. The canister shall be placed in an oven in a vertical, filling-end down position and maintained at 160 ± 5° F (71.1° C ± 2.8° C) temperature for the required time. Alternate method: the canister shall be placed in an oven at 207° F (97.2° to 100° C) in a vertical filling-end down position for the required time. When testing at 207° to 212° F (97.2° to 100° C) the temperature of the canisters shall be measured by thermocouple or thermometers in the burster wells of at least two canisters in each oven batch. Leakage shall be determined by visual examination.

5. PACKAGING

5.1 Packaging, interplant shipment. The canister assembly shall be packaged to provide adequate protection against deterioration and damage resulting from transit from the supply source to the first receiving activity.

5.2 Packing interplant shipment. The canister assembly, packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In conjunction with information furnished in the contract, shipping containers shall be marked in accordance with MIL-STD-129 as applicable.

6. NOTES

6.1 Intended use. This item is intended for use as a component of the M259 warhead and XM264 warhead.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

MIL-C-51449(EA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.3.3	2.75, WP Canister		D90-1-302

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY D90-1-328 PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:					
1	Leakage		100%	3.4	4.2.4.2/visual
2	WP Contamination		100%	3.3	4.2.4.2/visual
Major:					
101	Marking incorrect		0.40%	3.1.1	Visual
102	Filling incorrect		0.40%	3.2.2	4.2.4.1/Radiograph
103	Burster threads damaged		0.40%	3.1.1	Gage/visual
104	Workmanship		0.65%	3.1.1	Visual
Minor:					
201	Marking illegible or incorrect		0.65%	3.1.1	Visual
202	Damaged items		0.65%	3.1.1	Visual

NOTES:

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6.3 Caution. Because of the hazardous nature of WP operations, filling installations are advised to take all necessary precautions required by proper industrial hygiene and safety practices to protect personnel from injury while filling canisters in accordance with this specification.

6.4 Filling aid. Canisters filled with approximately 3.4 pounds of WP have been found to comply with 3.2.2. This information is furnished as an aid to procurement and not as a criterion for filling.

6.5 Decontamination. Canister may be decontaminated by passing them through a hot water and detergent spray washer or by flushing them with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well may be effected with a steam lance.

6.6 Supersession data. This specification supersedes Chemical Systems Laboratory Purchase Description EA-C-1159, 14 September 1977.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1340-A546

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-51449 31 JAN 1975

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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